

Attitude and Position Control for Stable Indoor Hover of a Parrot Mambo Minidrone

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Abstract—This report covers the foundation flight of a Parrot Mambo test series: holding a stable indoor hover with a Simulink flight controller deployed to the physical airframe. The control problem is set up from the vehicle's sensing and actuation reality: an underactuated airframe with six degrees of freedom, four motors, and state estimation fused from ultrasound, pressure, IMU, and optical-flow measurements. The attitude controller was iterated through six model versions before the deployed flight, alongside a separate position-control model. The final configuration holds a steady hover against drift and ground effect, and became the base layer every later flight in the series builds on.

Index Terms—Parrot Mambo, Simulink, attitude control, hover, PID, motor mixing, state estimation.

I. INTRODUCTION

Every capability demonstrated later in this flight series, from trajectory tracking to landing on a moving platform, sits on one prerequisite: the drone must first hold itself still in the air. This report documents that base layer, developed on the Parrot Mambo minidrone with the MathWorks Simulink toolchain.

The Parrot Mambo carries the sensor suite the whole series relies on: an ultrasound sensor timing pulses to the floor for altitude, a pressure sensor for altitude changes, an IMU providing linear acceleration and angular rates, and a downward-facing camera whose optical flow contributes horizontal velocity estimates. Four motors in an X configuration are commanded through motor mixing: collective thrust, a differential between the clockwise and counter-clockwise pairs for yaw, and side-to-side speed imbalances for pitch and roll. The airframe is underactuated, with six degrees of freedom and only four inputs, so every lateral motion begins as a deliberate tilt of the thrust vector.

II. SYSTEM ARCHITECTURE

All flight code was developed in Simulink against the Parrot Minidrone Support Package and deployed to the physical airframe, keeping the model structure the support package expects: an estimator consuming the sensor buses, a flight control system, and reference logic that the author modified per task. Structuring the work as small reusable subsystems made iteration and debugging on hardware considerably less painful.

The hover work involved two models the author developed and tuned: an attitude-control model, carried through six saved

iterations before the final version, and a position-control model layered on top of it. The attitude loop tracks roll, pitch, and yaw setpoints; the position loop turns position error into the small tilt commands the attitude loop executes.

III. METHOD

A. Control structure

Control follows the cascaded structure natural to a multirotor: position error maps to attitude references, attitude error maps through PID-style loops to torque demands, and motor mixing distributes those demands to the four rotors. Altitude is regulated on the fused ultrasound and pressure estimate.

B. Iteration

The attitude model went through six versions. Early versions exposed the usual gap between simulation behavior and a real airframe: gains that looked crisp in simulation produced oscillation or sluggish recovery on hardware, and were walked back toward a configuration that behaved on both.

IV. FLIGHT TESTING AND RESULTS

The deployed controller holds a steady indoor hover on the physical Mambo, on camera, without operator input during the hold. Station keeping remains within a small envelope around the takeoff point, with the drone visibly rejecting the small disturbances of indoor air and its own ground effect near the floor.

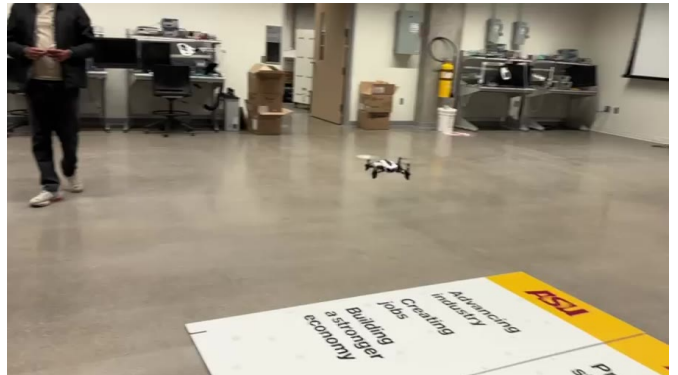


Fig. 1. The deployed hover flight: the Mambo holding station over the arena floor.

V. CHALLENGES

- Gains that survive the transfer from simulation to a real, imperfect airframe.
- Ground effect close to the floor disturbing the altitude loop.
- Small-airframe sensitivity: the Mambo's low inertia turns aggressive gains into visible oscillation immediately.

VI. EXTENSION: HEXACOPTER POSITION CONTROL IN SIMULATION

To test how far the quadrotor control concepts generalize, the author rebuilt the position-and-attitude control problem for a six-rotor airframe in a separate Simulink simulation study (Hexacopter_control_position_control). The cascaded architecture carries over unchanged: position error still maps to attitude references, and attitude error still maps to torque

demands. What changes is the allocation. Motor mixing must distribute thrust, roll, pitch, and yaw demands across three opposing rotor pairs instead of two, which alters the mixing matrix and introduces the actuation redundancy that quadrotors lack.

The takeaway was structural: the control architecture is airframe-agnostic, and porting it means re-deriving the allocation, not the loops. That redundancy is also the configuration's practical appeal, since a hexacopter retains attitude authority after losing a rotor, one reason the layout appears on inspection and heavier-lift platforms. The study remained simulation-only, complementing the hardware-deployed quadrotor work in this report.

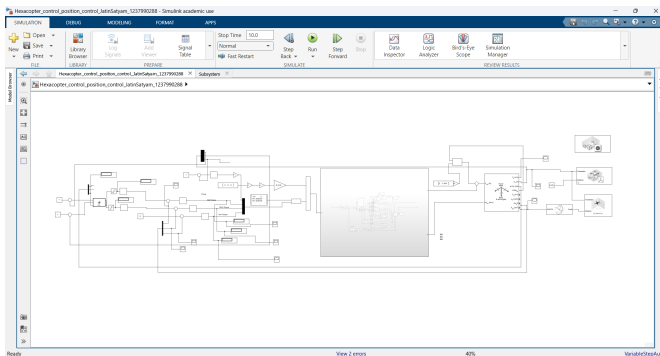


Fig. 2. The hexacopter position-and-attitude control model in Simulink.

VII. CONCLUSION

The hover controller delivered what the series needed: a trustworthy base layer. Six iterations of the attitude model produced a configuration stable enough to carry vision processing, trajectory tracking, and landing logic in the flights that followed.

APPENDIX: MATLAB FUNCTION CODE

The following MATLAB Function blocks are the author's custom code from the deployed Simulink model, reproduced verbatim.

Listing 1. Position error to attitude reference mapping (JS_Attitudecontrol_v6_FINAL.slx)

```
function [pitch, roll] = fcn(dx,dy,psi)
pitch=-dx*cos(psi)-dy*sin(psi);
roll=-dx*sin(psi)+dy*cos(psi);
end
```

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